

Data Wrangling

Group Project Guidelines

1. Overview

This project asks each group to take a real, self-sourced dataset through the full data wrangling pipeline covered in this course extraction, cleaning, transformation, and loading and then carry it forward into a machine learning model and a Power BI dashboard.

2. Group Formation

- Groups of 3. No individual or 3+ member submissions allowed.

3. Dataset Selection

Each group chooses its own dataset this is a deliberate part of the assessment. Choose something the group can genuinely reason about, not merely the first clean CSV found online.

3.1 Dataset requirements

- Real-world (not purely synthetic/toy) data, publicly available or ethically obtainable with permission.
- At least 1,000 rows and 8 columns, OR multiple related tables/files requiring joins — enough substance to require genuine wrangling.
- Must be "messy" enough to justify the pipeline: some mix of missing values, inconsistent formats, duplicates, outliers, or need for merging multiple sources. If the raw data is already clean, the group must argue why it still merits the full pipeline, or choose a messier source.
- No dataset already used verbatim in a course lab (e.g. `netflix_titles.csv`) pick something new.
- Must support both a meaningful ML task (a clear target/label for prediction, or natural grouping for clustering) and a dashboard-worthy set of dimensions/measures for Power BI check this before locking in a dataset.
- Acceptable sources: Kaggle, government/open-data portals, public APIs (e.g. weather, transit, finance), institutional repositories, or web-scraped data (with a note on scraping legality).

3.2 Approval checkpoint

Every group should discuss in advance about naming the dataset, source link, the question the group intends to answer, and why the dataset needs wrangling. Instructor sign-off is required before extraction work begins, to avoid late-stage rejections.

4. Required Pipeline & Deliverables

The project must visibly demonstrate each of the following stages, matching the course's Experiment structure.

4.1 Data Extraction

- Document how and from where the data was obtained (download, API call, scrape, or DB query).
- Note any access constraints, rate limits, authentication, or licensing terms encountered.

4.2 Data Cleaning

- Handle missing values, duplicates, type mismatches, and inconsistent categorical labels with justification for the method chosen (drop vs. impute vs. flag), not just the code.

- Maintain a short cleaning log: what was found, what was done, what was left as-is and why.

4.3 Data Transformation

- At least one non-trivial reshape or aggregation (pivot, group by, merge/join across tables, or time-based/recency aggregation).
- At least one engineered/derived feature that supports the analysis question.

4.4 Data Loading & Storage

- Load the final wrangled dataset into at least one persistent form beyond a flat CSV e.g. SQLite/SQL table, or partitioned/chunked export with validation.

4.5 Insights & Communication

- At least 3 concrete, non-obvious insights, each backed by a visualization or summary statistic.
- A short discussion of limitations what the data cannot tell you, and any wrangling decisions that could bias conclusions.

4.6 Machine Learning

- Frame at least one ML task on the wrangled dataset — classification, regression, clustering, or forecasting, whichever fits the data and question.
- Justify the model choice(s); compare at least two algorithms or one algorithm against a simple baseline.
- Follow proper methodology: train/test (or train/validation/test) split, appropriate preprocessing (scaling/encoding) fit only on training data, and cross-validation where feasible.
- Report evaluation metrics appropriate to the task (e.g. accuracy/precision/recall/F1/ROC-AUC for classification; RMSE/MAE/R² for regression) and interpret them honestly, including where the model underperforms.

4.7 Power BI Dashboard

- Build a Power BI (.pbix) dashboard connected to the wrangled dataset and, where relevant, model outputs (e.g. predicted vs. actual, cluster labels, forecast bands).
- Minimum 3 report pages/tabs: an overview page, a detail/drill-down page, and a model-insights page.
- Include at least one interactive element (slicer, filter, drill-through) and use visuals appropriate to each insight rather than defaulting to bar charts throughout.
- Dashboard should tell a coherent story for a non-technical stakeholder title, labels, and a brief narrative/annotation, not just raw charts.

5. Submission Package

1. Jupyter notebook — data wrangling and ML code, markdown commentary, and outputs, running end-to-end without errors.
2. Cleaning & transformation log — can be embedded in the notebook as markdown or submitted as a short separate document.
3. Power BI file (.pbix) — dashboard connected to final dataset/model outputs, plus a PDF export as a fallback.
4. Written report (5–7 pages) — objective, dataset description, methodology per pipeline stage, ML approach and results, dashboard walkthrough, insights, limitations. Word or PDF.
5. Presentation deck (8–10 slides) for the viva, including a live or screen-recorded dashboard demo.
6. Raw and cleaned dataset files (or a script to regenerate them, if the raw file is too large to submit).
7. Individual contribution statement — one line per member describing their primary responsibility.